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1 Step 12 — Sequence Models and LLM Agents in Strategic Settings: Experiment Report

Testbed: Kuhn Poker (exact Nash and exact exploitability from Step 02), with a narrow port to Leduc Hold'em as a complexity check. **Methods compared:** Nash-CFR · Behavioural Cloning · Decision Transformer · Adversarially Robust Decision Transformer (ARDT) · four LLM backends (offline stub, gpt-oss-20b, Qwen2.5-7B-Instruct, OpenThinker3-7B). **PhD connection:** Contribution #1 (behavioural adaptation), Contribution #2 (safe exploitation), Contribution #3 (evaluation methodology). **Scope of results:** all numbers are measured on real runs (RTX 5090, local models via LM Studio 0.4.20). Every figure is rendered from a committed results JSON.

Artifact caveat (read once). Two results were **retracted mid-session** and appear here only as retractions: an in-context-learning result of **+1.59** gap closed (impossible — see §Prediction vs reality, R5) and a Leduc LLM score of **-0.83** chips/hand produced by a faulty decoder (R6). The Nash reference differs by CFR budget — **0.0162** chips at 5,000 iterations (comparison table), **0.0061** at 50,000 (follow-on experiments); both are ≈ 0 . OpenThinker3's **plain-prompt** row is not comparable to its CoT row (34% of its probability mass goes to non-action tokens under a plain prompt).

How to read this report. The core build answers the raw step's five validation targets. The follow-on experiments were added because those five targets rank the methods without explaining them. Every claim is tied to a named artifact under `implementation/step12/implementation/`.

1.1 What this step is about

Two post-classical ways to play a game. **Sequence modelling** reframes RL as conditional sequence prediction: feed a GPT-style model (**return-to-go**, **state**, **action**) triples and predict the next action *conditioned on a desired future return* — no value function, no Bellman backup, no self-play. **ARDT** adds minimax return conditioning so the policy is robust to the worst-case opponent rather than tuned to whoever happened to be in the data. **LLM agents** skip training entirely: give a language model the rules in English and let it play.

Kuhn Poker is the shared testbed because its Nash equilibrium and exploitability are *exactly* computable, so every method gets a ground-truthed score rather than an anecdote. All methods are coerced into Step 02’s 12-info-set strategy format and measured with the same exact metric.

1.2 The headline comparison

Exploitability in chips (lower is closer to Nash); SMOKE profile, offline stub for the LLM rows.

Method	Exploitability (chips)	mbb/h	bluff(J)	value-bet(K)
Nash-CFR (reference)	0.0162	16.2	0.23	0.68
Behavioural Cloning	0.0550	55.0	0.20	0.68
ARDT	0.4691	469.1	0.33	0.35
Decision Transformer	0.7992	799.2	1.00	0.64
LLM (stub) × 3 prompt styles	0.8333	833.3	1.00	1.00

The two methods the step is about finish last. Plain behavioural cloning — the simplest possible baseline — lands within 0.04 chips of Nash and beats the return-conditioned Decision Transformer by roughly 14×. On near-Nash self-play data BC simply copies a near-Nash policy, whereas the DT must route that same policy through a return-conditioning channel that (as shown below) carries mostly luck. **Return conditioning actively destroys information here.**

1.2.1 Real models

Measured at temperature 0.7 with 24 samples per information set — a protocol point that matters, see *Measurement protocol* below.

Backend	style	expl. (chips)	bluff(J)	value- bet(K)	illegal	adapt
gpt-oss-20b	plain	0.2760	0.00	1.00	0%	+0.67
gpt-oss-20b	CoT	0.2500	0.08	1.00	1%	+0.79

Backend	style	expl. (chips)	bluff(J)	value- bet(K)	illegal	adapt
gpt-oss-20b	game- theory	0.3316	0.46	1.00	1%	+0.25
Qwen2.5- 7B	plain	0.3125	0.00	1.00	0%	+0.00
Qwen2.5- 7B	CoT	0.3183	0.58	1.00	0%	+0.42
Qwen2.5- 7B	game- theory	0.3032	0.50	1.00	0%	+0.21
OpenThinker3- 7B	CoT (n=12)	0.2882	0.33	1.00	16%	+0.92

A 7B model matches a 20B model: Kuhn rewards mixing at the right frequency, not knowledge or scale. Two failures are universal — every backend value-bets the King at 1.00 where Nash mixes at 0.68, and none bluffs the Jack under a plain prompt. LLMs get hand *ranking* right and *frequencies* wrong.

Reasoning tuning, isolated. OpenThinker3-7B is a reasoning-SFT of the *same* Qwen2.5-7B base, so the CoT rows isolate its effect: exploitability $0.318 \rightarrow 0.288$, bluff(J) $0.58 \rightarrow 0.33$ (against Nash’s 0.23), adaptation $+0.42 \rightarrow +0.92$ — bought with a **16% illegal-move rate** and **65× the tokens** (~6,500 vs ~99 per decision).

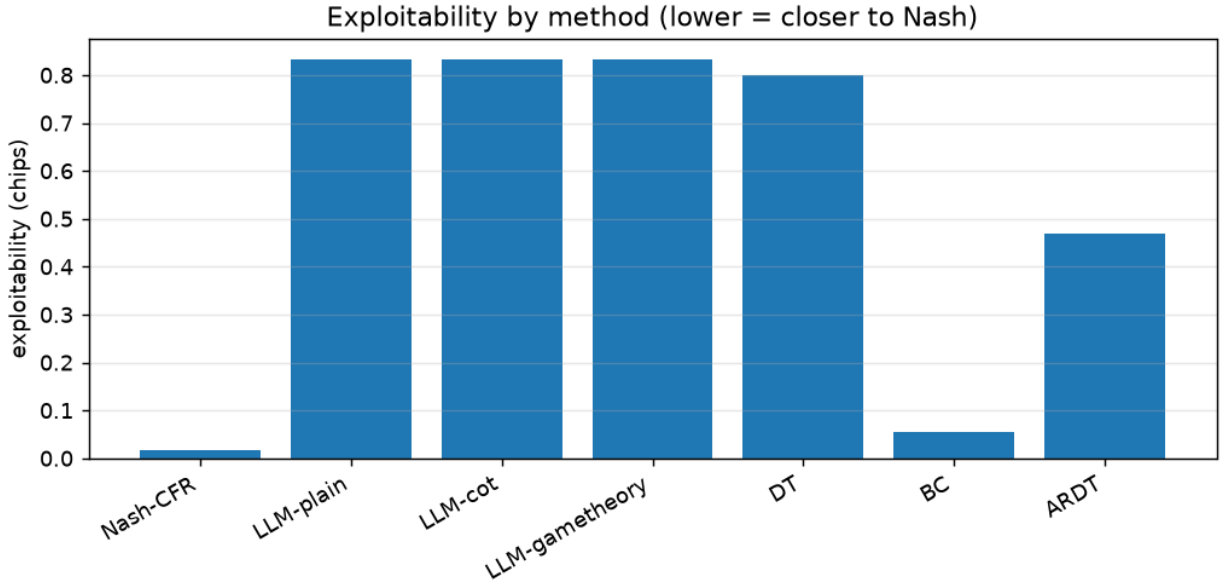


Figure 1: Exploitability by method on the exact Kuhn metric — plain behavioural cloning, included only as a contrast baseline, beats both of the methods this step is named after.

1.3 The five validation targets

`validate.py`, SMOKE profile: **3 PASS / 2 FAIL / 0 SKIP** (PyTorch and CUDA were live, so nothing was skipped).

#	Target	Measured	Verdict
1	DT: high return-to-go → lower exploitability than low	high(+2) = 0.6981 vs low(-2) = 0.6928 chips	FAIL
2	DT action varies by card (learns luck)	root P(bet) J/Q/K = 1.00/0.03/0.70, spread 0.97	PASS
3	ARDT < standard DT on the same mixed data	ARDT 0.5034 vs DT 0.7212 chips	PASS
4	ARDT within ~50 mbb/h of Nash	0.5034 chips vs a 0.05 tolerance	FAIL
5	LLM honestly more exploitable than Nash	0.8333 > 0.0162; real models 0.25-0.33	PASS

Both failures are genuine findings rather than tuning problems, and both are reconciled below.

Target #1 fails because the DT’s response to the conditioned return is **not ordered by how good that return is**. It does respond — a fine sweep of $\frac{1}{4}$ root P(**bet**) with the King runs $0.747 \rightarrow 0.020 \rightarrow 0.781$ — but with a sharp collapse at exactly $R = -1$, where it passes at **11 of 12** information sets and scores 1.98 chips. $R = -1$ is the modal return in the data (41.7% of steps) and the payoff of *folding*

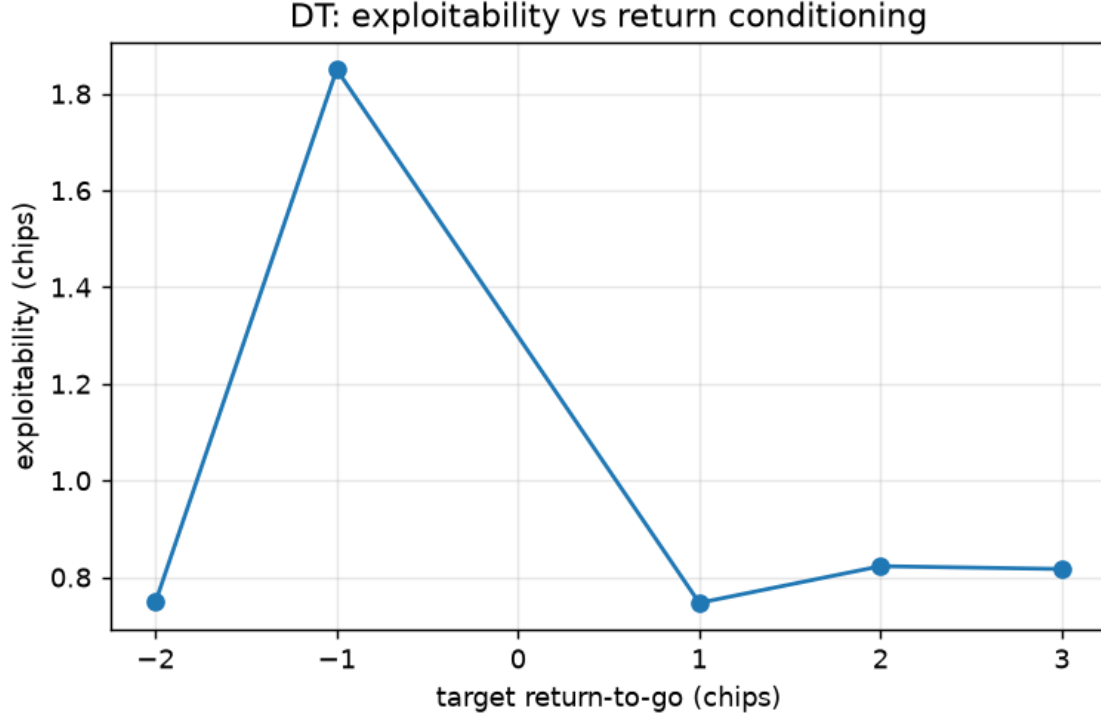


Figure 2: DT exploitability against the conditioned target return, including an impossible +3. The curve is flat across the real range with a sharp collapse at $R = -1$, the modal payoff and the payoff of folding.

1.4 MATH FLAG B — the ARDT expectile direction

The raw step writes $\tau = 0.9$ and calls it “pessimistic”. The implementation flagged this as inverted and defaulted to 0.1. **The paper confirms the flag:** ARDT’s Eq. (6) defines the expectile loss and Eq. (7) states $\lim_{\alpha \rightarrow 0} g_{\alpha} = \min$, $\lim_{\alpha \rightarrow 1} g_{\alpha} = \max$, and Algorithm 1 line 1 runs $\alpha = 0.01$. This was independently corroborated by the module’s own self-test, which recovers the *analytic* expectiles of a skewed sample ($\tau=0.1 \rightarrow -1.951$, $\tau=0.9 \rightarrow 0.000$).

The empirical sweep contradicts the theory, and the reason is instructive. The relabel target moves monotonically with τ ($-0.626 \rightarrow +0.854$), confirming the mechanism is wired correctly — yet exploitability is **lowest at $\tau = 0.9$** , the optimistic side. Algorithm 1 line 7 explains it: ARDT relabels with $R_t = Q_v(s_t, a_t)$, a state-**action** value from two coupled networks (Eqs. 8–11), whereas this implementation relabels with a state-only $V(s)$. A state-only target cannot distinguish “this state is bad” from “*this action* is bad” — exactly the discrimination ARDT depends on — so pushing $\tau \rightarrow 0$ hands the DT a uniformly negative number, which selects the *folding* line rather than the *robust* one. EXPECTILE_TAU is deliberately left at 0.1, the theoretically correct side; changing it to 0.9 to obtain a better number would be rigging the result.

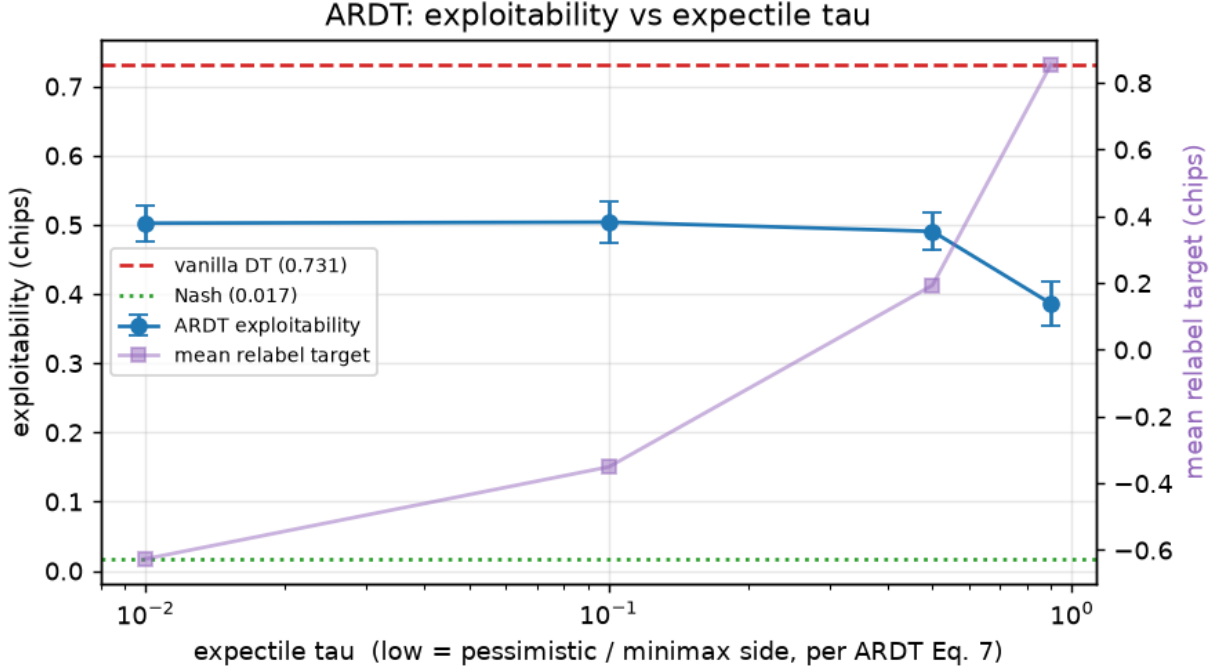


Figure 3: ARDT exploitability against the expectile tau, with the mean relabel target on the right axis. The relabel target moves monotonically as theory requires, but exploitability is lowest on the optimistic side.

1.5 Follow-on experiments: what the scalar score hides

The five targets rank the methods but do not explain them. Six further experiments were run.

1.5.1 Exact strategy extraction, 24× cheaper

Reading the action distribution from **token logprobs** (prefill "Action:", sum probability mass over surface forms) replaces 288 sampled calls with **12**, with zero sampling variance. Validated against sampling on identical prompts: mean $|P(\text{bet})|$ gap **0.027** against a binomial SE of 0.102 (consistent). It is *not* valid with a chain-of-thought prompt — the prefill contradicts “reason first”, giving an out-of-distribution policy — a boundary documented in the code.

1.5.2 Where the loss actually is

Information set	P(bet)	Nash	deviation	share of leak
2p (Queen, opponent checked)	1.000	0.000	1.000	41.4%
2b (Queen, facing a bet)	0.253	0.347	0.094	13.1%
3 (King, root)	1.000	0.561	0.439	0.1%

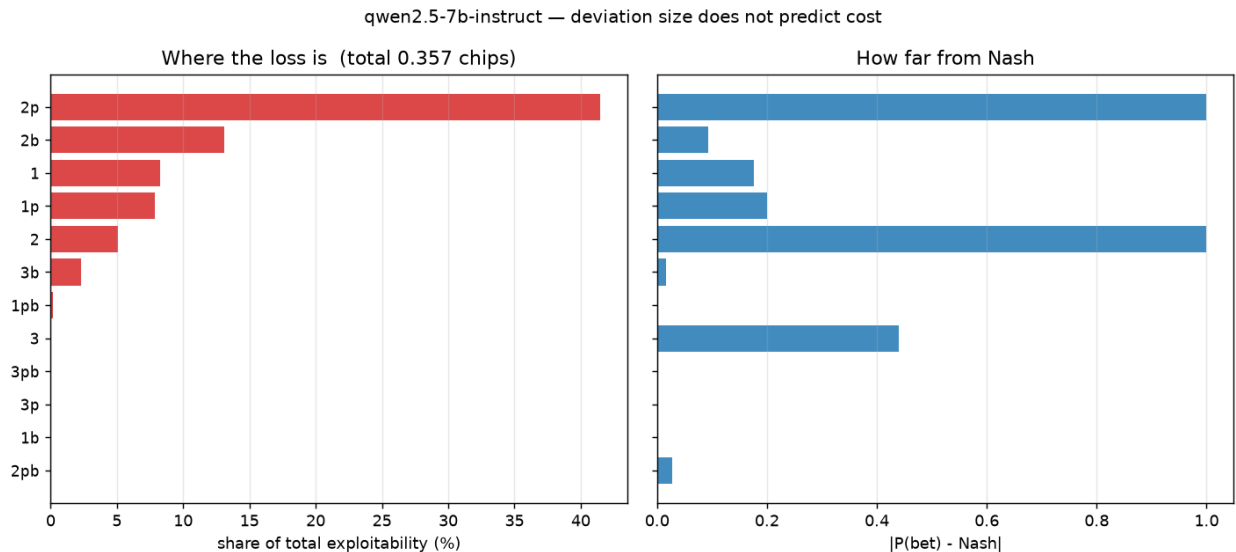


Figure 4: Each information set’s share of total exploitability, beside how far it deviates from Nash. The two orderings barely correspond.

This **corrects a headline stated earlier in this step**. “Every LLM value-bets the King at 1.00 where Nash mixes at 0.68” was reported as a signature failure — it costs **0.1%** of the leak. Over-betting a hand that is never behind is nearly free. The damage is one Queen node: always betting the Queen after the opponent checks is **41.4%** of total exploitability, and repairing that single decision alone moves the model from 0.357 to 0.209 chips. **Deviation magnitude and cost are nearly uncorrelated**, which is precisely what a single number hides.

1.5.3 The models play better than they can explain

Asked, at the same information sets and with identical situation text, what percentage of the time they should bet:

	MAE vs Nash (stated)	MAE vs Nash (executed)	expl. if it played what it says	expl. as played
Qwen2.5-7B	0.353	0.246	0.921 chips	0.357
gpt-oss-20b	0.434	0.328	1.576 chips	0.392

Both models’ **stated** strategies are substantially worse than their **played** ones — 2.6× and 4.0× more exploitable respectively. The failure modes differ (Qwen answers “50%” at 11 of 12 information sets; gpt-oss answers ~0% at 8 of 12, including all three King nodes where Nash bets 100%), which strengthens the shared conclusion: **extracting a strategy from an LLM by asking it yields something worse than probing its behaviour**.

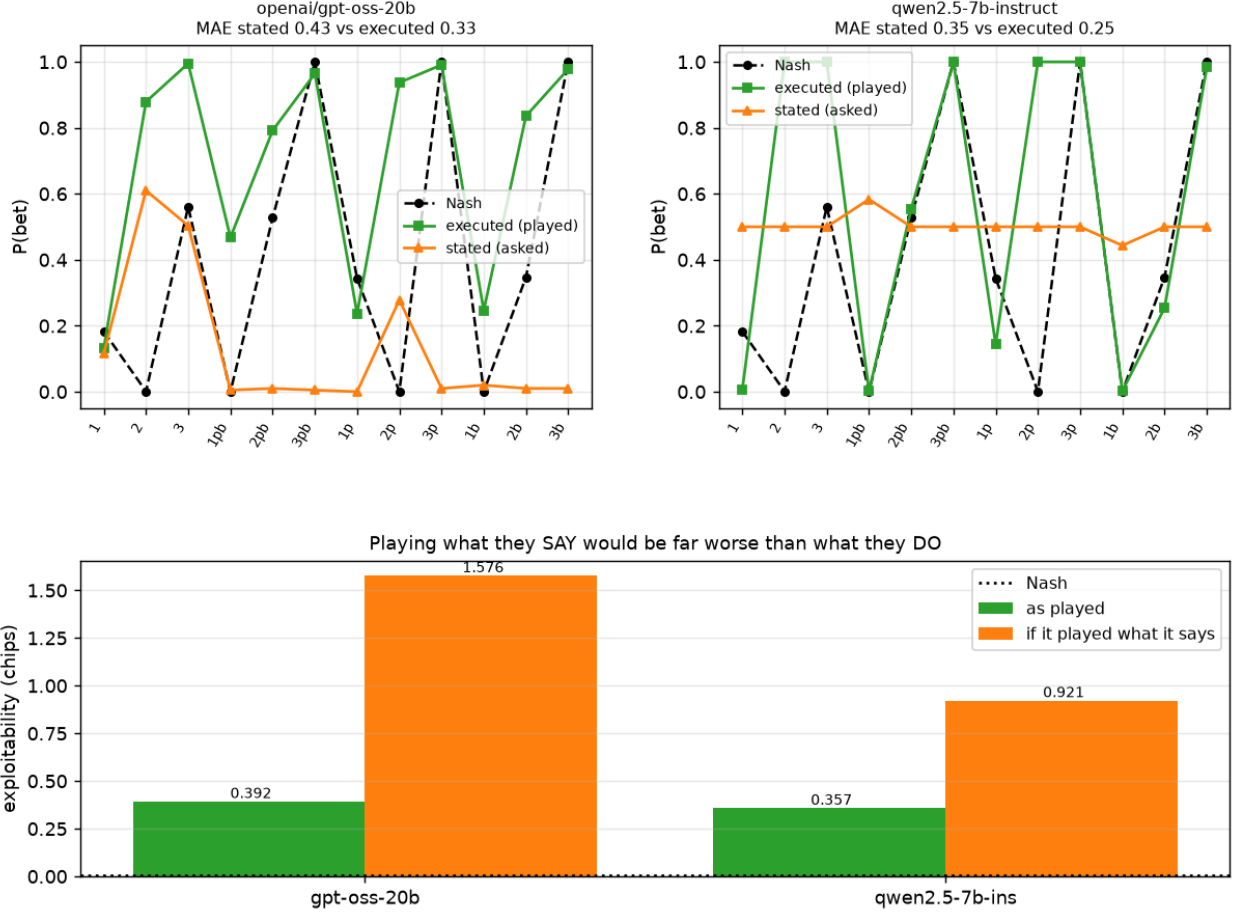


Figure 5: Stated versus executed betting frequency at each information set for both models, with Nash for reference, and the exploitability of each model’s played strategy against a strategy built from its own stated frequencies.

1.5.4 Exploitative by default, not adaptive

Sessions of 120 hands against a fixed archetype, with the observed history of previous hands in context. Baselines are computed **exactly** on the agent’s own realised deals, so neither deal luck nor opponent randomness enters the comparison.

Opponent	Nash	best-response		gap closed	learning (2nd – 1st half)
		ceiling	agent $\pm$ SE		
AlwaysPass	+0.229	+0.981	+0.850 $\pm$ 0.056	+0.83 $\pm$ 0.07	+0.00
AlwaysBet	+0.319	+0.530	+0.383 $\pm$ 0.169	+0.31 $\pm$ 0.80	−0.52
TightPassive	+0.088	+0.316	+0.092 $\pm$ 0.122	+0.02 $\pm$ 0.53	−0.15

Mean learning **−0.22**. Against the one well-powered cell the agent captures 83% of the available exploitation **from the first half onward and never improves** — a fixed loose-aggressive prior, not opponent modelling.

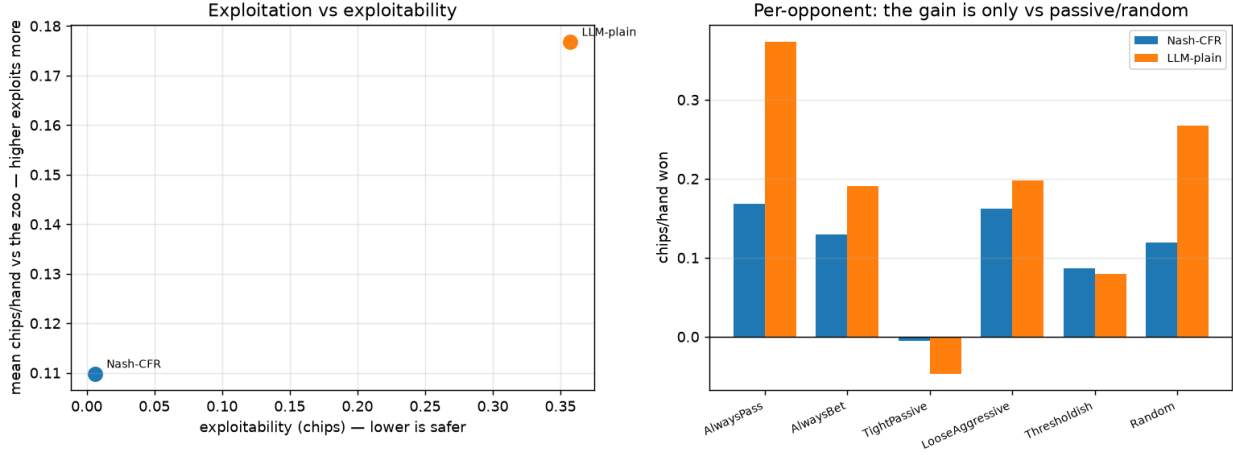


Figure 6: Exploitation against exploitability, and the per-opponent breakdown showing the gain is confined to passive and random opponents.

The exploitation/exploitability frontier corroborates it: the LLM **exploits 61% harder than Nash while being 59× more exploitable** (0.177 vs 0.110 chips/hand), but the entire gain is against *passive or random* opponents (AlwaysPass 0.374 vs 0.168). Against the two most competent archetypes it is **worse** than Nash. This is the safe-exploitation trade-off of Contribution #2, measured.

1.5.5 Head-to-head

20,000 hands per pair, seats alternated. Nash versus Nash is 0.0000 exactly and Nash loses to nobody, which validates the tournament.

Row player	vs Nash	vs gpt-oss	vs Open-		mean	exploitability
			Thinker3	vs Qwen		
Nash-CFR	0.0000	+0.1061	+0.1245	+0.0806	+0.104	0.0061
Qwen2.5-7B	−0.0806	+0.1618	+0.1817	0.0000	+0.088	0.3571
gpt-oss-20b	−0.1061	0.0000	+0.1535	−0.1618	−0.038	0.3917
OpenThinker3-7B	−0.1245	−0.1535	0.0000	−0.1817	−0.153	0.8940

The **7B beats the 20B**, and the ordering is strictly transitive and matches the exploitability ordering exactly — in this population, exploitability predicts head-to-head results.

1.6 Leduc Hold'em — a complexity check

A narrow port (encoder, dataset, DT training, and a return-conditioning sweep scored by chips/hand against a near-Nash opponent) plus an LLM scouting pass. Leduc has **936** information sets versus Kuhn’s 12, **15** distinct payoff values versus 4, two betting streets and a board card.

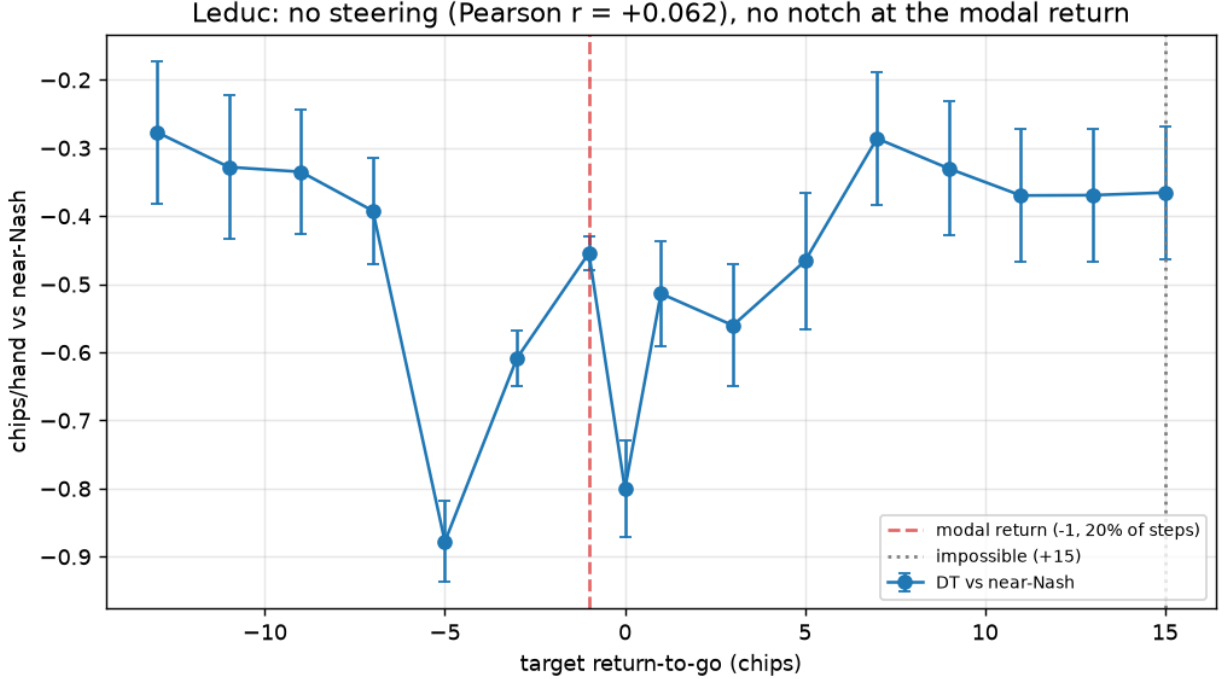


Figure 7: DT performance against the conditioned target return on Leduc, with standard errors. There is no collapse at the modal return and no monotone trend.

The Kuhn notch does not reproduce — at the modal return the DT is 0.2 SE from the mean of the other targets. **But conditioning still does not steer:** Pearson $r = +0.062$, Spearman $\rho = -0.054$. It is not inert (the spread across targets is 0.60 chips/hand, many times the per-point SE), it is simply not ordered by target quality. The impossible target saturates, the one Kuhn prediction that reproduces cleanly.

LLM competence degrades sharply. On Leduc the LLM scores -0.463 ± 0.132 chips/hand against near-Nash, statistically indistinguishable from the DT’s -0.454 ; against the exploitable zoo it manages -0.071 where Nash makes $+0.582$, losing even to **Random**. Only **31–54%** of the 936 information sets were ever reached, so reach probability makes full enumeration unnecessary.

Its 23.4% illegal-action intent is **one systematic misconception**, not diffuse confusion: **FOLD_WHEN_FREE** is **100%** of the illegal mass, while **RAISE_AT_CAP** is 0.0000 and non-action (format failure) is 0.0001. It is concentrated in round 2 with nothing due (0.5138, up to 0.99), on weak unpaired hands against a high board — the model conflates “my hand is weak” with “I should fold”, forgetting that checking is free. The near-zero round-1 rate shows this is

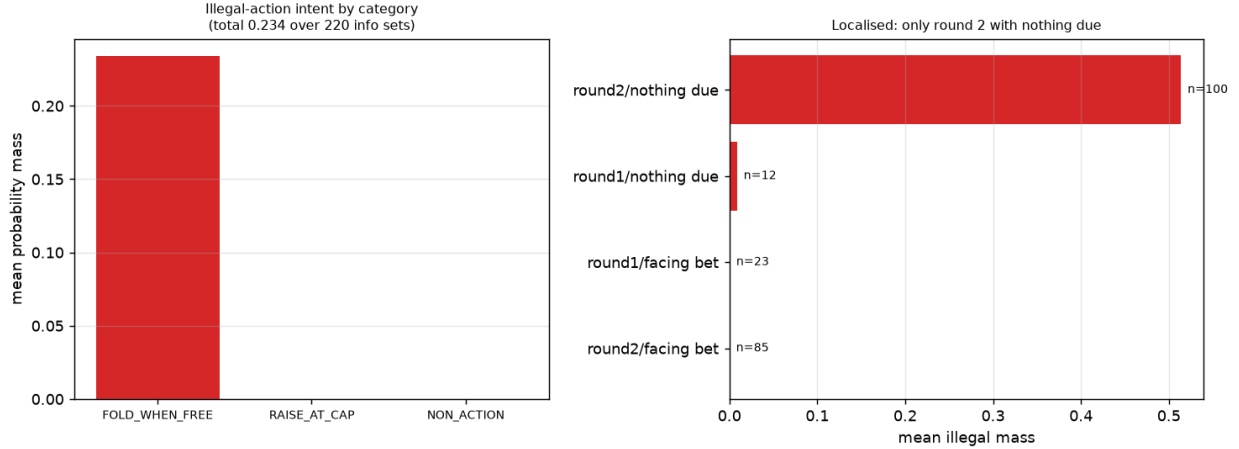


Figure 8: Illegal-action intent by category and by situation. A single category accounts for all of it, concentrated in one situation.

board-texture-triggered rather than a rules gap, making it a prompting finding rather than a capability ceiling.

1.7 Prediction vs reality reconciliation

Pre-run predictions are kept; what actually happened is appended (WORKFLOW §0.1).

R1 — Return conditioning never steers, and the proposed mechanism was half right. The Kuhn explanation (magnitude encodes the betting line, sign is card luck, so the modal fold payoff selects “the folding line”) predicted the effect should weaken with a richer payoff alphabet. Leduc removed the notch exactly as predicted **but steering did not appear in its place**. The mechanism explains the *notch*, not the *failure*, which survives 15 payoff values, two streets and a board card.

R2 — MATH FLAG B was correct about τ and understated the structural gap. Confirmed against Eqs. 6–7; the real gap is $V(s)$ versus the paper’s $Q(s, a)$ (Algorithm 1 line 7). Target #4 stays red.

R3 — Behavioural cloning wins, and the King “failure” was nearly free. See *The headline comparison* and *Where the loss actually is*.

R4 — The going-in hypothesis was refuted. “Knows the frequency but cannot sample it” predicted stated $\approx$ Nash and executed $\neq$ Nash. Measured: stated is *worse* than executed on both models.

R5 — An in-context-learning result of +1.59 was retracted. A gap closed above 1.0 means beating the exact best response, which is impossible; the ceiling was verified by hand ($K \rightarrow +2$, $Q \rightarrow 0$, $J \rightarrow -1 \Rightarrow +0.333$, matching the computed +0.3387), identifying the agent’s figure as the artefact.

Cause: a 60-hand session compared against baselines sampled over *different* deals, where Kuhn’s per-hand standard deviation of ~ 1.2 gives an SE comparable to the entire Nash→BR span. With exact paired baselines the conclusion **reverses** to no learning.

R6 — A Leduc LLM score of -0.83 was retracted. The tokenizer splits RAISE into " RA" and FOLD into " F" while CALL/CHECK stay whole, so whole-word matching discarded 70% of the probability mass and *inverted* the policy — a model raising at $p=0.953$ was scored as calling $\sim 97\%$ of the time. Corrected with prefix-aware matching to -0.463 ± 0.132 .

R7 — LLM mixing cannot be measured at temperature 0. A real model plays a *pure* strategy there, so every measured frequency degenerates to 0 or 1: bluff(J) read 0.75, 0.25 and 1.00 across three runs of the same configuration, and raising the sample count from 4 to 24 did not help. At temperature 0.7 the frequencies become intermediate and the illegal-move rate becomes nonzero.

1.8 Trustworthiness and sample adequacy

Four results in this step looked like discoveries and were measurement artefacts (R5, R6, a seat-0 return that was 1.7 SE of sampling noise, and a Leduc trend that vanished at $20\times$ the hands). Each was caught by a cheap consistency check against something exactly computable — an exact game value, an exact best-response ceiling, a standard error, or a probability-mass conservation diagnostic.

Separately, `plotting.py` originally contained **only** a self-test that plotted hard-coded numbers into the results directory, and it was the sole figure-producing path in the step; following the runbook literally would have committed a fabricated figure. It was replaced with a renderer that reads exclusively from committed JSON, which is how every figure in this report was produced.

Sample adequacy: head-to-head cells use 20,000 hands; zoo matches 4,000; the Leduc sweep 4,000 per target. The opponent-modelling cells for AlwaysBet and TightPassive are **underpowered** (SE ± 0.80 and ± 0.53 , because their Nash→BR spans are narrow) and only the AlwaysPass cell is conclusive.

Training variance in the neural rows. The dataset, CFR and the offline stub are seeded and reproduce exactly, but each invocation of the comparison table **retrains** the neural models, so DT/BC/ARDT carry run-to-run variance. Measured across two runs of an identical configuration, DT exploitability moved **$0.671 \rightarrow 0.799$** (a $\sim 19\%$ swing) while the two deterministic rows (Nash-CFR 0.0162, LLM-stub 0.8333) reproduced to four decimal places. The numbers quoted here are those of the committed artifact `results/comparison_SMOKE_stub.json`; the qualitative ordering is unaffected, but **these rows should not be compared at three decimal places across runs.**

1.9 Limitations (ranked by how much they affect the conclusions)

1. **Every Leduc LLM conclusion rests on one model, one prompt style, 600 hands** (SE ± 0.13). Enough to say the Kuhn advantage does not transfer; not enough to rank models on Leduc.
2. **ARDT is a documented simplification of the published method** — single-sided, state-only $V(s)$ instead of the coupled state-action $Q(s, a)$. Targets #3–4 test the proxy, not ARDT.
3. **No exact Leduc exploitability.** Leduc’s metric lives in Step 03, whose package is also named `cfr` and collides with Step 02’s; chips/hand against near-Nash was used instead.
4. **Kuhn results are SMOKE-profile** (5,000 trajectories, 5,000 CFR iterations, small networks). The qualitative ordering is robust, but absolute values would shift at scale.
5. **Logprob extraction is validated for plain prompts only**; CoT figures come from sampling.
6. **OpenThinker3 was measured on the CoT row only** ($n=12$); a full pass is ~ 9 hours at ~ 38 s per decision.
7. **B5’s two underpowered cells** (above) — tightening them needs roughly $30\times$ more hands.

1.10 Conclusions and research directions

Conclusions. Return conditioning does not steer a Decision Transformer in poker, on either of two games, and the reason is not the size of the payoff alphabet: in a zero-sum imperfect-information game the realised return is dominated by factors the agent does not control. Plain behavioural cloning beats both of the methods this step is named after. LLMs get hand ranking right and mixing frequencies wrong, cannot verbalise the strategy they actually play, do not learn opponents from observed play, and lose their apparent competence when the game grows by one street.

Research directions (each tied to a measured effect):

- *Replace ARDT’s state-only relabeling with the paper’s coupled state-action estimators* (Eqs. 8–11 plus the Algorithm-1 warm-up). This is the named, evidence-backed reason the proxy underperforms and the change to make before Step 13’s fixed logs.
- *Treat return-conditioned DT as the wrong instrument for offline poker* — the value of ARDT’s relabeling is precisely that it swaps an uncontrollable conditioning target for a controllable one.
- *Report per-decision decompositions rather than a scalar* in the Step 14 evaluation framework; deviation magnitude is not a proxy for cost.
- *Probe behaviour rather than asking* in any LLM-based opponent-modelling component.
- *Test the one-line Leduc prompt fix* (“you may check for free”) before drawing any conclusion about LLM capability at multi-street poker.

1.11 Reproduction

from the repository root, with the venv active

`cd implementation/step12/implementation`

```
python validate.py                                # 5 validation targets -> 3 PASS / 2 FAIL / 0 SKIP
python comparison_table.py                        # headline table -> results/comparison_SMOKE_<backend>
python tau_sweep.py --seeds 3                    # MATH FLAG B, empirical half
python exploitability_decomposition.py --style plain --method logprob
python frequency_elicitation.py --repeats 3 --temp 0.7 --exec-style plain
python opponent_modeling.py --hands 120 --style cot
python exploitation_vs_zoo.py --style plain --hands 4000
python llm_head_to_head.py --tournament --hands 20000
python leduc_stage0.py                           # Leduc DT return-conditioning sweep
python leduc_llm.py --hands 600                  # Leduc LLM scouting
python leduc_illegal_taxonomy.py                 # illegal-action breakdown
python plotting.py                               # all 12 figures, from committed JSON only
```

Real-model runs require a local OpenAI-compatible server. Protocol that matters: **load exactly one model** (`lms unload --all` then `lms load <model>`), restart the server after any killed run (orphaned connections hold all parallel slots and later runs hang indefinitely), and set `STEP12_LLM_TEMP=0.7` — at temperature 0 the measured frequencies are meaningless (R7).

Determinism: datasets, CFR and network initialisation are seeded; the offline LLM stub is deterministic, so the SMOKE table and validation harness reproduce exactly without a GPU or network. Real-model numbers are **not** bit-reproducible (MoE routing under batched serving), which is why frequency estimates use 24 samples and why the temperature protocol is stated explicitly.